

1) Vektorový prostor $V = \mathbb{C}^n$ se standardním skalárním součinem a označme

$$u_j = a_j + ib_j, v_j = x_j + iy_j; a_j, b_j, x_j, y_j \in \mathbb{R}.$$

$$\text{Potom platí: } \langle u, v \rangle = \sum_{j=1}^n [(a_j - ib_j)(x_j + iy_j)] = \sum_{j=1}^n (a_j x_j + i a_j y_j - i b_j x_j + b_j y_j)$$

$$\text{tedy: } \operatorname{Re} \langle u, v \rangle = \sum_{j=1}^n (a_j x_j + b_j y_j); \operatorname{Im} \langle u, v \rangle = \sum_{j=1}^n (a_j y_j - b_j x_j)$$

$$\frac{1}{4} (\|u+v\|^2 - \|u-v\|^2) = \cancel{\frac{1}{4} (\|u+v\|^2 - \|u-v\|^2)}$$

$$= \frac{1}{4} \left(\sum_{j=1}^n |u_j + v_j|^2 - \sum_{j=1}^n |u_j - v_j|^2 \right) = \frac{1}{4} \left(\sum_{j=1}^n ((a_j + x_j)^2 + (b_j + y_j)^2 - (a_j - x_j)^2 - (b_j - y_j)^2) \right) =$$

$$= \frac{1}{4} \sum_{j=1}^n (a_j^2 + 2a_j x_j + x_j^2 + b_j^2 + 2b_j y_j + y_j^2 - a_j^2 + 2a_j x_j - x_j^2 - b_j^2 + 2b_j y_j - y_j^2) =$$

$$= \frac{1}{4} \sum_{j=1}^n (4a_j x_j + 4b_j y_j) = \sum_{j=1}^n (a_j x_j + b_j y_j) = \operatorname{Re} \langle u, v \rangle$$

$$\frac{1}{4} (\|u - iv\|^2 - \|u + iv\|^2) = \frac{1}{4} \sum_{j=1}^n ((a_j + y_j)^2 + (b_j - x_j)^2 - (a_j - y_j)^2 - (b_j + x_j)^2) =$$

$$= \frac{1}{4} \sum_{j=1}^n (a_j^2 + 2a_j y_j + y_j^2 + b_j^2 - 2b_j x_j + x_j^2 - a_j^2 + 2a_j y_j - y_j^2 - b_j^2 - 2b_j x_j - x_j^2) =$$

$$= \frac{1}{4} \sum_{j=1}^n (4a_j y_j - 4b_j x_j) = \sum_{j=1}^n (a_j y_j - b_j x_j) = \operatorname{Im} \langle u, v \rangle$$